TYPES OF GENE THERAPY

- Ex vivo gene therapy:- transfer of genes to cultured cells and reinsertion.

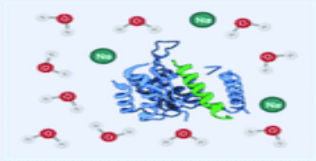
- In vivo gene therapy:- direct delivery of genes into the cells of a particular tissue in the body

Exploratory

a)

In silico

Computational modeling



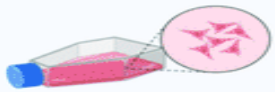
Mechanisms prediction.
Molecular interactions.
Effects elucidation.

b)

In vitro

2D models
Cell monolayers

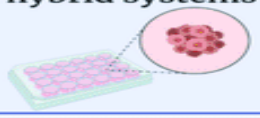
Conventional culture



Cell-to-cell interactions.
Simple manipulation.
Inexpensive.
Single microenvironment.

3D models
Spheroids
Organoids

Scaffold-free, scalffold-based and hybrid systems

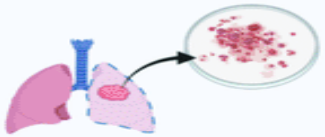


Cell-to-cell and cell-to-matrix interactions.
Mimicking intratumor heterogenity (eg. hypoxic core).

c)

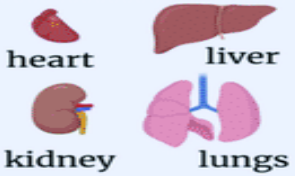
Ex vivo

Tissue explants
Surgically resected tumor



To preserve the structure, multicellular 3D architecture and cancer signaling.

Physiological models
Isolated tissues and organs



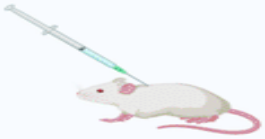
Tissue or organ function evaluation.
Elucidation of mechanisms.

Confirmatory

d)

In vivo

Murine models
Genetically engineered animals.
Chemical induction.
Transplantable.



To study cancer onset, invasion and metastasis. It represents a significant step between in vitro systems and clinical assays.

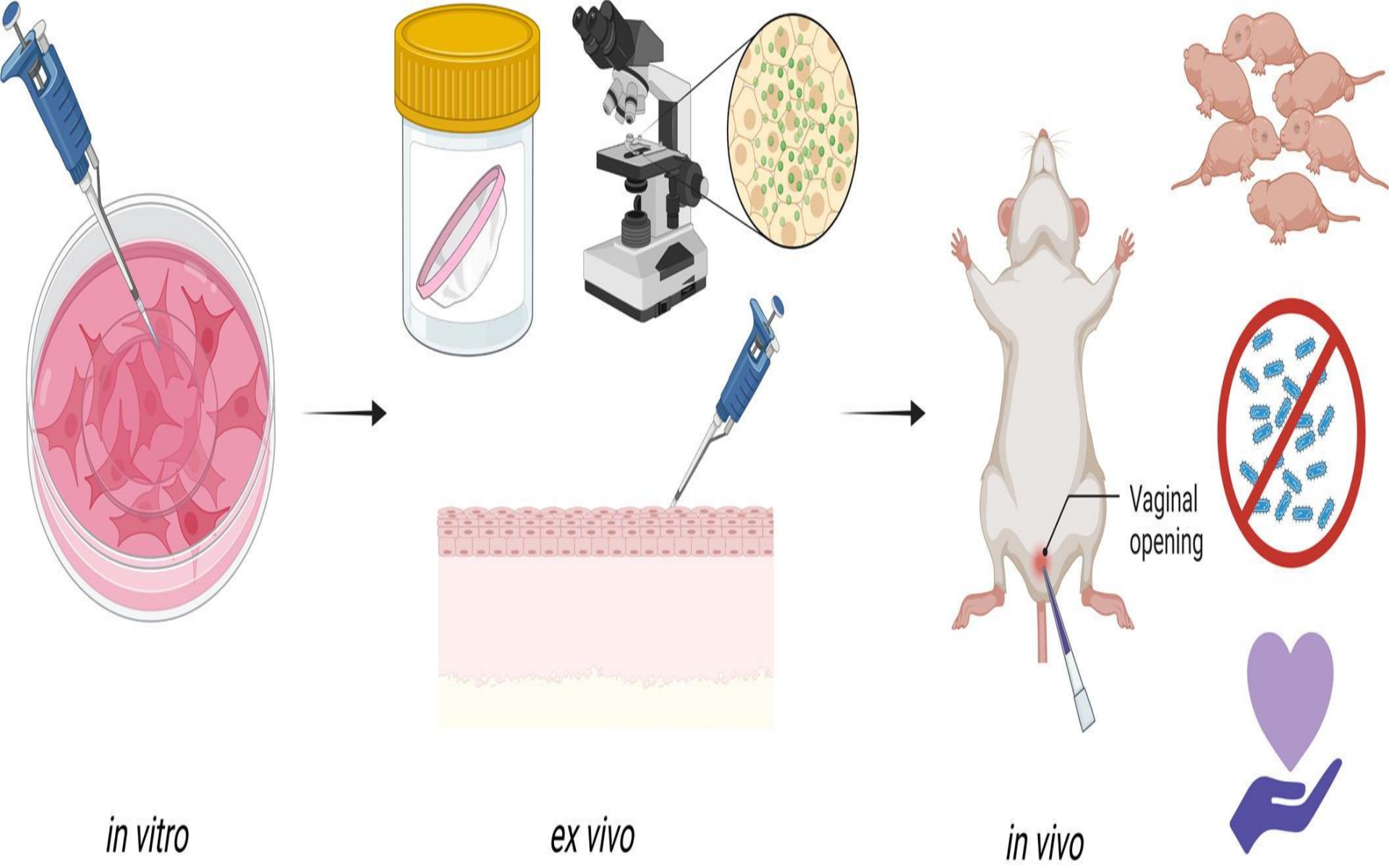
C. elegans

***D. rerio* (zebrafish)**



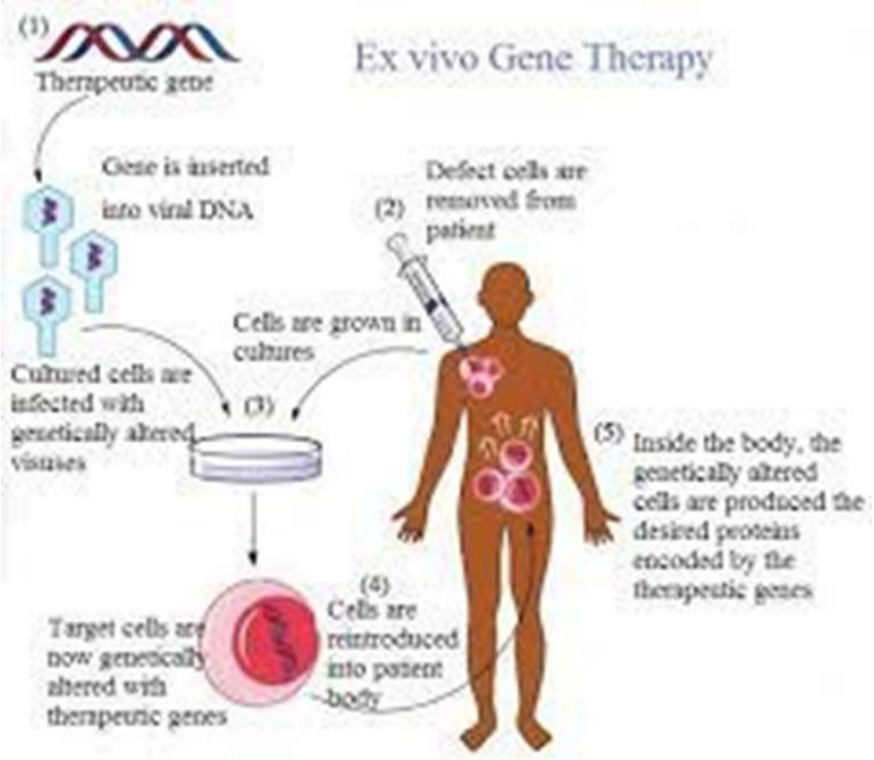
Contributes to the development of new therapeutic interventions.

Non-vertebrate and non-mammalian animal model widely used to study developmental biology and cancer


Vaginal
opening
in vitro
ex vivo
in vivo
Created with BioRender

EX VIVO GENE THERAPY

- Isolate cells with genetic defect from a patient.
- Grow the cells in culture.
- Introduce the therapeutic genes .
- Select genetically corrected cells and grow.
- Transplant the modified cells to the patient.

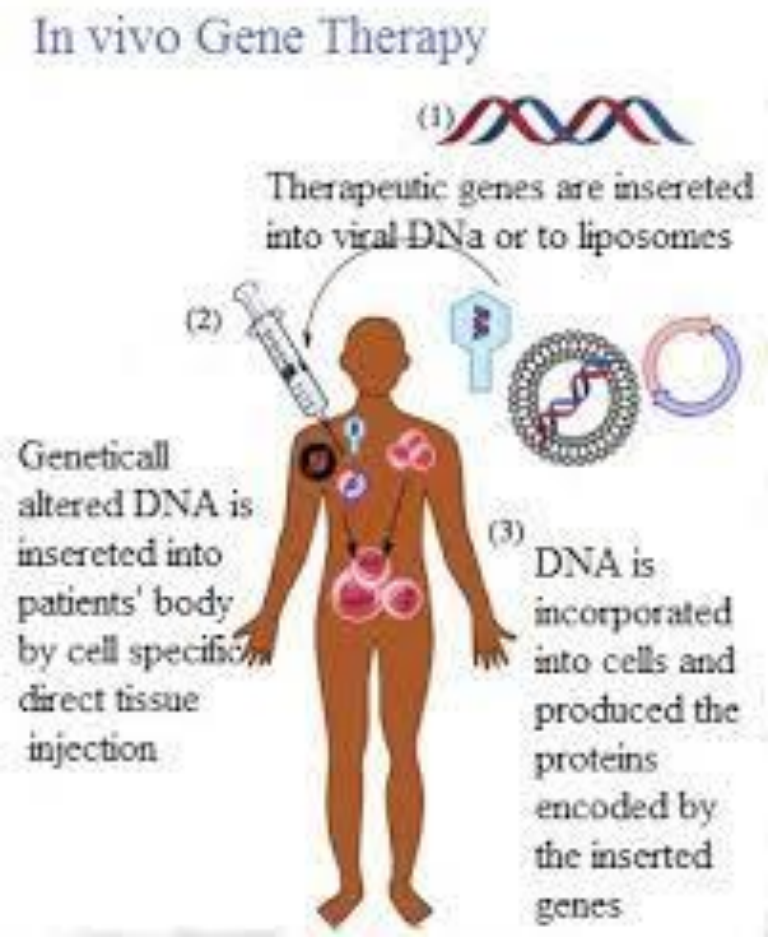


EXAMPLE OF EX VIVO GENE THERAPY

- 1st gene therapy – to correct deficiency of enzyme, **Adenosine deaminase** (ADA).
- Performed on a 4yr old girl Ashanthi DeSilva.
- Was suffering from SCID- **Severe Combined Immunodeficiency**.
- Caused due to defect in gene coding for ADA.
- Deoxy adenosine accumulate and destroys T lymphocytes.
- Disrupts immunity , suffer from infectious diseases and die at young age.

IN VIVO GENE THERAPY

- Direct delivery of therapeutic gene into target cell into patients body.
- Carried out by viral or non viral vector systems.
- It can be the only possible option in patients where individual cells cannot be cultured in vitro in sufficient numbers (e.g. brain cells).
- In vivo gene transfer is necessary when cultured cells cannot be rei-mplanted in patients effectively.



EXAMPLE OF IN VIVO GENE THERAPY

- **Therapy for cystic fibrosis**
- In patients with cystic fibrosis, a protein called **cystic fibrosis transmembrane regulator** (CFTR) is absent due to a gene defect.
- In the absence of CFTR **chloride ions** concentrate within the cells and it draws water from surrounding.
- This leads to the accumulation of sticky mucous in respiratory tract and lungs.
- Treated by in vivo replacement of defective gene by adenovirus vector .

VECTORS IN GENE THERAPY

- To transfer the desired gene into a target cell, a carrier is required. Such vehicles of gene delivery are known as **vectors**.

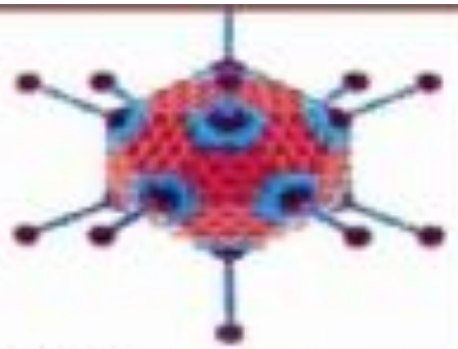
- 2 main classes
  - **Viral** vectors
  - **Non viral** vectors

Ideal vector

- TARGET the right cells
- INTEGRATE the gene in the cells.
- ACTIVATE the gene.
- AVOID harmful side effects.
- No universal vector exists.



VIRAL VECTORS



- Viruses introduce their genetic material into the host cell as part of their replication cycle.
- remove the viral DNA and using the virus as a vehicle to deliver the therapeutic DNA.
- The viruses used are altered to make them safe, although some risks still exist with gene therapy.

Types of viral vectors

- A number of viruses have been used for human gene therapy, including :

1. Retrovirus
2. Adenovirus
3. Adeno-associated virus
4. Herpes simplex virus

I. Viral vectors

Retroviruses, adenoviruses and adeno-associated viruses (AAV) some commonly used viral vectors whereas some less commonly used viral vectors are derived from the Herpes simplex virus (HSV-1), the baculovirus etc. Viral vectors include:

A. Adenoviral vectors

Adenoviruses are viruses that carry their genetic material in the form of double-stranded DNA. They cause respiratory, intestinal, and eye infections in humans (especially the common cold). When these viruses infect a host cell, they introduce their DNA molecule into the host. The genetic material of the adenoviruses is not incorporated (transient) into the host cell's genetic material. The DNA molecule is left free in the nucleus of the host cell, and the instructions in this extra DNA molecule are transcribed just like any other gene. The only difference is that these extra genes are not replicated when the cell is about to undergo cell division, so the descendants of that cell will not have the extra gene.

As a result, treatment with the adenovirus will require re-administration in a growing cell population. This vector system has been promoted for treating cancer and indeed the first gene therapy product to be licensed to treat cancer, **Gendicine**, is an adenovirus. Gendicine, an adenoviral p53-based gene therapy was approved by the Chinese food and drug regulators in 2003 for treatment of head and neck cancer. **Advexin**, a similar gene therapy approach, was turned down by the US Food and Drug Administration (FDA) in 2008.

B. Adeno- Associated Virus (AAV)

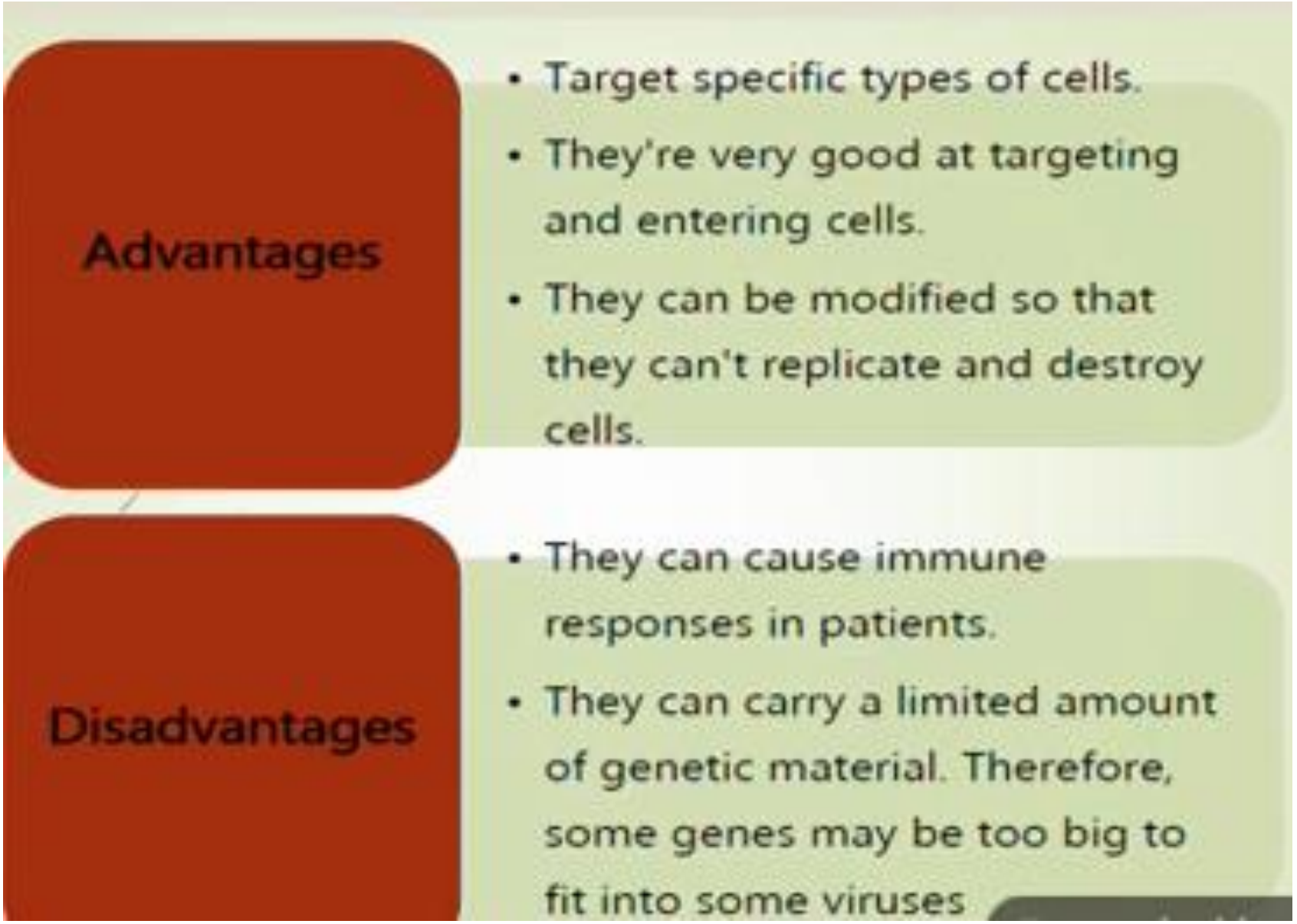
Adeno-associated viruses (AAVs) are a group of small, single-stranded DNA viruses which cannot usually undergo productive infection without co-infection by a helper virus, such as an adenovirus. The insert size for AAV is 4.5 kb, with the advantage of long-term gene expression as they integrate into chromosomal DNA. AAVs are highly safe as the recombinant adeno associated vectors contains only gene of interest and 96% viral genes are deleted.

C. Retroviruses vectors

The genetic material in retroviruses is in the form of RNA molecules, while the genetic material of their hosts is in the form of DNA. When a retrovirus infects a host cell, it will introduce its RNA together with some enzymes, namely **reverse transcriptase** and **integrase**, into the cell. This RNA molecule from the retrovirus must produce a DNA copy from its RNA molecule before it can be integrated into the genetic material of the host cell. The process of producing a DNA copy from an RNA molecule is termed **reverse transcription**. It is carried out by one of the enzymes carried in the virus, called reverse transcriptase. After this, DNA copy is produced and is free in the nucleus of the host cell, it must be incorporated into the genome of the host cell. This process is done by another enzyme carried in the virus called integrase

Now that the genetic material of the virus has been inserted, it can be said that the host cell has been modified to contain new genes. If this host cell divides later, its descendants will all contain the new genes.

One of the problems of gene therapy using retroviruses is that the integrase enzyme can insert the genetic material of the virus into any arbitrary position in the genome of the host; it randomly inserts the genetic material into a chromosome. If genetic material happens to be inserted in the middle of one of the original genes of the host cell, this gene will be disrupted (insertional mutagenesis). If the gene happens to be one regulating cell division, uncontrolled cell division (i.e., cancer) can occur.


Advantages
• Target specific types of cells.
• They're very good at targeting and entering cells.
• They can be modified so that they can't replicate and destroy cells.
Disadvantages
• They can cause immune responses in patients.
• They can carry a limited amount of genetic material. Therefore, some genes may be too big to fit into some viruses